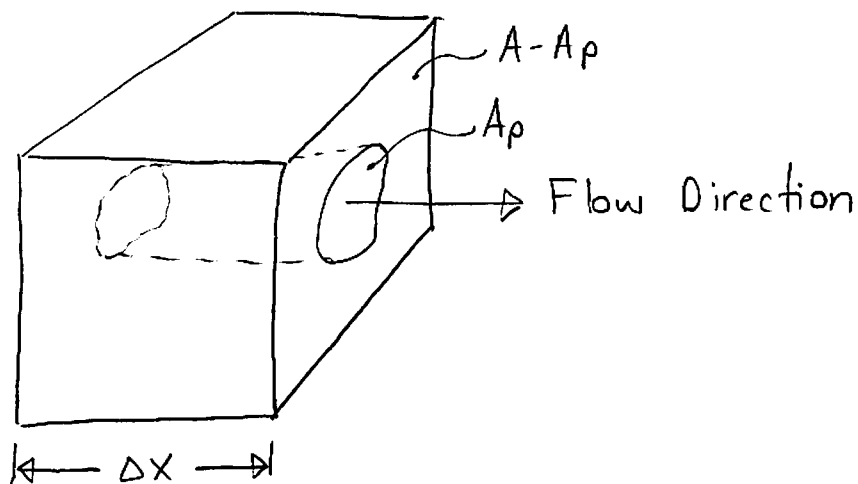


Heat Transfer in Porous Media

Let's consider a simple 1-D model in order to obtain our energy equation in porous media:



According to our previous definition: $\phi = \frac{A_p \Delta x}{A \Delta x}$ ①

For the solid domain:

$$\rho_s c_s \frac{\partial T}{\partial t} = k_s \frac{\partial^2 T}{\partial x^2} + \overset{\substack{\nearrow \text{heat generation/volume}}}{q_s'''} \Rightarrow \rho_s, c_s, k_s \Rightarrow \text{solid properties}$$

If $T = \text{constant}$ over the solid volume, we can integrate volume-
trically

$$\Delta x (A - A_p) \rho_s c_s \frac{\partial T}{\partial t} = \Delta x (A - A_p) k_s \frac{\partial^2 T}{\partial x^2} + \Delta x (A - A_p) q_s''' \quad ②$$

For the fluid domain:

$$\rho_f c_{Af} \left(\frac{\partial T}{\partial t} + u_p \frac{\partial T}{\partial x} \right) = k_f \frac{\partial^2 T}{\partial x^2} + \mu \Phi \quad ③$$

Note we neglected our compressibility term $(\beta T \frac{\partial P}{\partial t}) \Rightarrow \text{incompressibl.}$

Note, T here is the same in the fluid as the solid.
Local thermal equilibrium. Not always true though so be
carefull when using it.

For example $\Rightarrow$ nuclear reactors or electronics cooling: $T_f \neq T_s$.
 If no Local Equilibrium, we must solve the fluid & solid coupled equations.
 Integrating eq. ③ over the pore volume: Note: $Au = \iint_{A_p} u_p dA_p$

$$\Delta x A_p \rho_f c_{pf} \frac{\partial T}{\partial t} + \Delta x A_p \rho_f c_{pf} u \frac{\partial T}{\partial x} = \Delta x A_p k_f \frac{\partial^2 T}{\partial x^2} + \Delta x u \iint_{A_p} \Phi dA_p \quad ④$$

Our last term represents the internal heating due to viscous dissipation.

Since this is a loss term ($E_{out} = E_{in}$), then the last term is equal to the mechanical power required to drive the flow.

We can use fluid mechanics to show that:

$$\underbrace{\Delta x u \iint_{A_p} \Phi dA_p}_{\text{Viscous heating}} = \underbrace{Au \left(-\frac{\partial P}{\partial x} + \rho_f g_x \right) \Delta x}_{\text{Mechanical work input}} \quad ⑤$$

Back substituting ⑤ into ④ and adding ② $\Rightarrow$ divide out by $A \Delta x$

$$\boxed{\left[\phi \rho_f c_{pf} + (1-\phi) \rho_s c_s \right] \frac{\partial T}{\partial t} + \rho_f c_{pf} u \frac{\partial T}{\partial x} = \left[\phi k_f + (1-\phi) k_s \right] \frac{\partial^2 T}{\partial x^2} + (1-\phi) \dot{q}_s''' + u \left(-\frac{\partial P}{\partial x} + \rho_f g_x \right)}$$

$\hookrightarrow$ Porous media energy equation (incompressible)

$$\boxed{k = \phi k_f + (1-\phi) k_s} \Rightarrow \text{porous medium thermal conduct.}$$

Note, since we added ② & ④, this assumes a parallel model, hence k is valid for this assumption. In general, k needs to be experimentally measured.

We can define a capacity ratio of our medium as:

$$\sigma = \frac{\phi \rho_f c_{pf} + (1-\phi) \rho_s c_s}{\rho_f c_{pf}}$$

$\Rightarrow$ capacity ratio
Doesn't mean much, more for notation as we will see.

With this new notation, we can write: Note: $q''' = (1-\phi) q_s'''$

$$\rho_f c_{pf} \left(\sigma \frac{\partial T}{\partial t} + u \frac{\partial T}{\partial x} \right) = k \frac{\partial^2 T}{\partial x^2} + q''' + \frac{\mu}{K} u^2$$

In general for 3D:

$$\rho_f c_{pf} \left(\sigma \frac{\partial T}{\partial t} + \nabla \cdot \nabla T \right) = k \nabla^2 T + q''' + \frac{\mu}{K} (\nabla)^2$$

$\hookrightarrow \nabla = (u, v, w) \equiv$ volume averaged velocity vector

If heat generation & viscous heating are negligible:

$$\sigma \frac{\partial T}{\partial t} + u \frac{\partial T}{\partial x} + v \frac{\partial T}{\partial y} + w \frac{\partial T}{\partial z} = \alpha \left(\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} + \frac{\partial^2 T}{\partial z^2} \right)$$

Where:

$$\alpha = \frac{k}{\rho_f c_{pf}} \equiv \text{homogeneous porous medium thermal diff.}$$

Note: Our main assumptions here are:

- 1) Homogeneous medium: solid material & fluid is distributed evenly in the medium.
- 2) Isotropic medium: i.e. K & $k \neq f(\text{direction})$. Note, if not the case, then:

$$u = \frac{K_x}{\mu} \left(-\frac{\partial p}{\partial x} + \rho g_x \right), \quad v = \frac{K_y}{\mu} \left(-\frac{\partial p}{\partial y} + \rho g_y \right), \quad w = \frac{K_z}{\mu} \left(-\frac{\partial p}{\partial z} + \rho g_z \right)$$

$$\sigma \frac{\partial T}{\partial t} + u \frac{\partial T}{\partial x} + v \frac{\partial T}{\partial y} + w \frac{\partial T}{\partial z} = \alpha_x \frac{\partial^2 T}{\partial x^2} + \alpha_y \frac{\partial^2 T}{\partial y^2} + \alpha_z \frac{\partial^2 T}{\partial z^2}$$

- 3) At any point in the medium, the solid & fluid are in local thermal equilibrium.
- 4) Local $Re = \frac{\phi u k^{1/2}}{\mu} < 10$ (Darcy's Law is Applicable)
Laminar flow.

Non-LTE Heat Transfer

If the fluid and solid are not in thermal equilibrium, we must solve a coupled set of equations:

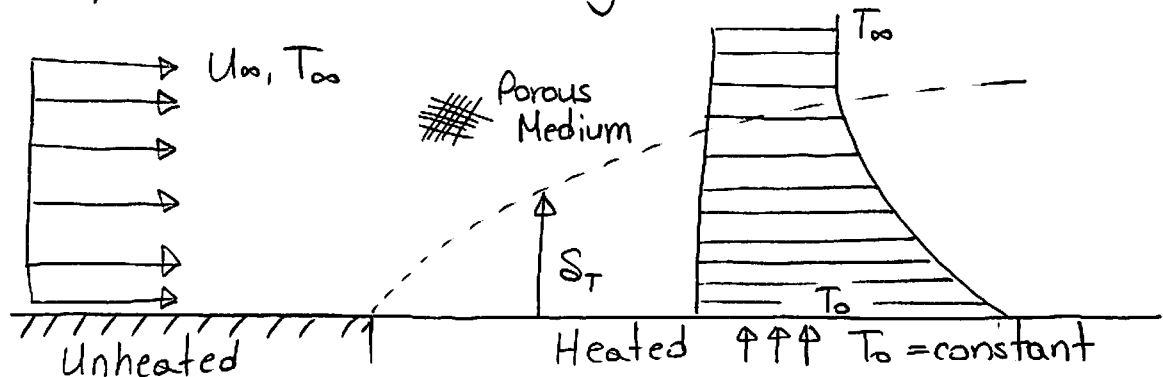
$$(1-\phi) \rho_s c_s \frac{\partial T_s}{\partial t} = (1-\phi) k_s \frac{\partial^2 T_s}{\partial x^2} + h (T_f - T_s) \quad (1)$$

$$\phi \rho_f c_f \frac{\partial T_f}{\partial t} + \rho_f c_f u \frac{\partial T}{\partial x} = \phi k_f \frac{\partial^2 T_f}{\partial x^2} - h (T_f - T_s) \quad (2)$$

Need to solve ① and ② simultaneously. Note, this example still assumes incompressible, viscous heating is negligible, and zero volumetric heating.

Boundary Layers

Let's say we had the following situation (incompressible flow)



Our governing equations become:

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0 \quad (\text{mass conservation})$$

$$u = -\frac{K}{\mu} \frac{\partial P}{\partial x} \quad ; \quad v = -\frac{K}{\mu} \frac{\partial P}{\partial y}$$

Considering that the b.l. is slender; with uniform parallel flow:

$$u = U_{\infty}, \quad v = 0, \quad p(x) = -\frac{\mu}{K} U_{\infty} x + \text{const}$$

↳ Solved by integrating Darcy's eqn.

Applying a scaling analysis: $u \frac{\partial T}{\partial x} \sim \alpha \frac{\partial^2 T}{\partial y^2}$ (energy eqn.)

$$\begin{aligned} u &\sim U_{\infty} & x &\sim x \\ \partial T &\sim \Delta T & y &\sim \delta_T \\ \partial^2 T &\sim \Delta T \end{aligned}$$

$$U_{\infty} \frac{\Delta T}{x} \sim \alpha \frac{\Delta T}{\delta_T^2} \Rightarrow \boxed{\frac{\delta_T}{x} \sim Pe^{-1/2}}$$

$$\boxed{Nu_x = h \frac{x}{K} \sim \frac{x}{\delta_T} \sim Pe^{1/2}} \quad \rightarrow \quad \boxed{Pe = \frac{U_{\infty} x}{\alpha}} \equiv \text{Peclet number}$$

Note, $\alpha \equiv$ thermal diffusivity of the porous medium $= \frac{k}{\rho c_p}$

Note also the similarity to the Blasius - Pohlhausen solution for a flat plate. Very similar however no hydrodynamic boundary layer development due to the porous media.

Same as $Pr \ll 1$ case where $u \sim U_{\infty}$ (see pg. (42) of notes). Only difference is before, we used fluid properties, & here we use porous media properties.

Similarity Solution ($T_0 = \text{constant} \equiv$ wall temperature)

Assuming the following similarity variable:

$$\eta = \frac{y}{x} Pe_x^{1/2}, \quad \theta(\eta) = \frac{T - T_0}{T_{\infty} - T_0}$$

Converting our energy equation

$$\left. \begin{aligned} 2T &= (T_\infty - T_0) 2\theta \\ 2^2 T &= (T_\infty - T_0) 2^2 \theta \\ U &= U_\infty \end{aligned} \right\} \text{Back substitute into energy equation}$$

$$U \frac{\partial T}{\partial x} = \alpha \frac{\partial^2 T}{\partial y^2} \Leftrightarrow U_\infty \frac{\partial \theta}{\partial x} (T_\infty - T_0) = \alpha \frac{\partial^2 \theta}{\partial y^2} (T_\infty - T_0)$$

Now we need to convert from $x \rightarrow \eta$

$$\frac{\partial \theta}{\partial x} = \frac{\partial \theta}{\partial \eta} \cdot \frac{\partial \eta}{\partial x} \Rightarrow \frac{\partial \eta}{\partial x} = \frac{\partial}{\partial x} \left(\frac{y}{x} \text{Pe}_x^{1/2} \right) = -\frac{y}{2} \left(\frac{U_\infty}{\alpha x^3} \right)^{1/2}$$

$$\frac{\partial \theta}{\partial x} = -\frac{y}{2} \left(\frac{U_\infty}{\alpha x^3} \right)^{1/2} \theta' \quad (1)$$

Now let's deal with the right hand side:

$$\frac{\partial^2 \theta}{\partial y^2} = \frac{\partial}{\partial y} \left(\frac{\partial \theta}{\partial \eta} \cdot \frac{\partial \eta}{\partial y} \right) = \frac{\partial}{\partial \eta} \left(\frac{\partial \theta}{\partial \eta} \cdot \frac{\partial \eta}{\partial y} \right) \cdot \frac{\partial \eta}{\partial y} = \frac{\partial^2 \theta}{\partial \eta^2} \left(\frac{\partial \eta}{\partial y} \right)^2$$

$$\frac{\partial \eta}{\partial y} = \left(\frac{U_\infty}{\alpha x} \right)^{1/2} \Rightarrow \left(\frac{\partial \eta}{\partial y} \right)^2 = \frac{U_\infty}{\alpha x}$$

$$\frac{\partial^2 \theta}{\partial y^2} = \frac{U_\infty}{\alpha x} \theta'' \quad (2)$$

Putting (1) & (2) together:

$$-U_\infty \left(\frac{y}{2} \left(\frac{U_\infty}{\alpha x^3} \right)^{1/2} \right) \theta' = \cancel{\alpha} \frac{U_\infty}{\cancel{\alpha} x} \theta''$$

$$-\frac{1}{2} \frac{y}{x} \left(\frac{U_\infty x}{\alpha} \right)^{1/2} \theta' = \theta''$$

$$\boxed{\theta'' + \frac{1}{2} \eta \theta' = 0} \Rightarrow \text{ODE} \Rightarrow \text{Solvable by separation of variables}$$

B.C.'s $\Rightarrow \theta(0) = 0$
 $\theta(\eta \rightarrow \infty) = 1$

Solving, we obtain:

$$\boxed{\theta(\eta) = \text{erf}\left(\frac{\eta}{2}\right)}$$

So for heat transfer: $q'' = -k \frac{\partial T}{\partial y} \Big|_{y=0}$

$$q'' = -k(T_\infty - T_0) \left. \frac{\partial \theta}{\partial y} \right|_{y=0} = -k(T_\infty - T_0) \left. \frac{\partial \theta}{\partial \eta} \right|_{\eta=0} \cdot \left. \frac{\partial \eta}{\partial y} \right|_{y=0}$$

$$\frac{\partial \eta}{\partial y} = \left(\frac{U_\infty}{\alpha x} \right)^{1/2} ; \quad \left. \frac{\partial \theta}{\partial \eta} \right|_{\eta=0} = \pi^{-1/2} \quad (\text{from tables or Wolfram})$$

$$\boxed{q'' = -k(T_\infty - T_0) \pi^{-1/2} \cdot \left(\frac{U_\infty}{\alpha x} \right)^{1/2}} \Rightarrow \text{Wall heat flux}$$

$$\Rightarrow T_0 = \text{constant}, Re_x < 10$$

For a wall heating a fluid, $T_0 > T_\infty$

$$Nu_x = \frac{q''}{T_0 - T_\infty} \cdot \frac{x}{k} = \frac{\cancel{k} \pi^{-1/2} \left(\frac{U_\infty}{\alpha x} \right)^{1/2} \cdot x}{\cancel{k}} = \left(\frac{U_\infty x}{\pi \alpha} \right)^{1/2}$$

$$\boxed{Nu_x = 0.564 Pe_x^{1/2}} \Rightarrow T_0 = \text{constant}, Re_x < 10, \text{Incompressible}$$

Note, in scaling analysis we had $Nu_x \sim Pe^{1/2}$ (Good!)

$$\boxed{\overline{Nu} = \frac{\bar{h} L}{k} = 1.128 Pe_L^{1/2}}$$

Constant Heat Flux

For the constant heat flux case, we can do another similarity analysis, but this time with:

$$\left. \begin{aligned} \frac{\partial T}{\partial y} &= -\frac{q''}{k} \quad \text{at } y=0 \\ T &\rightarrow T_\infty \quad \text{at } y \rightarrow \infty \end{aligned} \right\} \text{B.C.'s}$$

$$\zeta = y \left(\frac{U_\infty}{\alpha x} \right)^{1/2} ; \quad \tau(\zeta) = \frac{T(x, y) - T_\infty}{(q''/k) (\alpha x / U_\infty)^{1/2}}$$

Doing the math & converting our energy PDE into an ODE we obtain:

$$\tau'' + \frac{1}{2}(\zeta \tau' - \tau) = 0$$

$$\left. \begin{aligned} \tau'(0) &= -1 \\ \tau(\zeta \rightarrow \infty) &= 0 \end{aligned} \right\} \text{B.C.'s}$$

Differentiating once, we can do separation of variables

$$\frac{2}{23} \left(\tau'' + \frac{1}{2} (3\tau' - \tau) \right) = 0$$

$$\tau''' + \frac{1}{2} \cancel{\tau'} + \frac{1}{2} 3\tau'' - \frac{1}{2} \cancel{\tau} = 0$$

$$\tau''' + \frac{1}{2} 3\tau'' = 0 \Rightarrow \frac{2}{23} (\ln(\tau'')) = \frac{1}{\tau''} \cdot \frac{2}{23} (\tau'') = \frac{\tau'''}{\tau''}$$

$$\boxed{\frac{\tau'''}{\tau''} = -\frac{1}{2} 3} \Rightarrow \text{ODE} \Rightarrow \text{Similar to Blasius: } \frac{f'''}{f''} = -\frac{1}{2} f$$

To solve, we integrate 3 times in a row and then apply our B.C.'s: (For a similar integral, see pg. (49a) of notes)

$$\boxed{\tau(\xi) = \frac{2}{\pi^{1/2}} \exp\left(-\frac{\xi^2}{4}\right) - 3 \operatorname{erfc}\left(\frac{\xi}{2}\right)} \Rightarrow q'' = \text{const} \Rightarrow Re_k < 10$$

Solving for our Nusselt number (I skipped a few steps here for the sake of brevity):

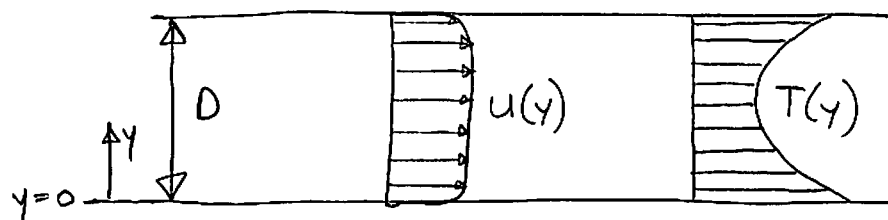
$$\boxed{Nu_x = \frac{q'' x}{k [T_o(x) - T_\infty]} = 0.866 Pe_x^{1/2}} \Rightarrow \begin{aligned} &\Rightarrow \text{Incompressible} \\ &\Rightarrow q'' = \text{constant} \\ &\Rightarrow Re_k < 10 \end{aligned}$$

$$\boxed{\overline{Nu} = \frac{q'' L}{k [\overline{T_o} - T_\infty]} = 1.329 Pe_L^{1/2}} \Rightarrow \begin{aligned} &\Rightarrow \text{Averaged Nusselt \#} \\ &\Rightarrow \text{Remember, } k, \alpha \text{ are porous media properties} \end{aligned}$$

Internal Flows filled with Porous Media

For the case of fully developed flow of a fluid through a porous medium that fills a long channel, we can use our Darcy flow model $\Rightarrow$ volume averaged velocity u is uniform across the whole channel.

This indicates what we call "slug flow"



⇒ Different from previous analysis where $u(y) \neq \text{const.}$
see pg. (127) of notes

Note, our bulk or mean temperature reduces to :

$$T_m = \frac{1}{\bar{u}A} \cdot \int_A uT dA \Rightarrow \text{since } u(y) = \bar{u} = \text{constant} \neq f(y)$$

$$= \frac{\bar{u}}{\bar{u}A} \int_A T dA$$

$$\boxed{T_m = \frac{1}{A} \int_A T dA} \Rightarrow \text{Mean temperature for slug flow.}$$

⇒ $A = \text{channel cross section.}$

To solve, we can use our previous approach of solving the energy equation for channel or pipe flows with $u = u_\infty = \text{constant} = \frac{k}{u} \left(-\frac{\partial P}{\partial x} \right)$

I've skipped the derivations but good to do for homework to check your understanding:

For Tubes (Internal Diameter = D)

$$\boxed{Nu_D = \frac{q''(x)}{T_o - T_m(x)} \cdot \frac{D}{k} = 5.78} \quad (\text{tube, } T_o = \text{constant})$$

$$\boxed{Nu_D = \frac{q''}{T_o(x) - T_m(x)} \cdot \frac{D}{k} = 8} \quad (\text{tube, } q'' = \text{constant})$$

For channels (Parallel Plates, spacing = D)

$$\boxed{Nu_D = \frac{q''(x)}{T_o - T_m(x)} \cdot \frac{D}{k} = 4.93} \quad (\text{parallel plates, } T_o = \text{constant})$$

$$\boxed{Nu_D = 6} \quad (\text{parallel plates, } q'' = \text{constant})$$